

B.Sc.CSIT 4th Semester

Operating System: CSC259

Lab-01: FAMILIARIZATION WITH LINUX AND GCC COMPILER

OBJECTIVES

- TO LEARN BASIC LINUX COMMANDS
- TO CREATE AND MANIPULATE FILE
- TO COMPILE AND RUN USING GCC COMPILER

LINUX BASIC COMMANDS

Some of the basic linux commands are:

- a) **pwd** : print working directory, it will print path to your current directory
- b) **ls**: provide list of names of files and directories
- c) **cd** : Change your current directory eg: cd desktop
- d) **mkdir** : to create a new directory. eg: mkdirsoch
- e) **rm -d** : delete the directory. eg: rm -rf soch (for kali linux OS) rm -d soch (for ubuntu)

FILE MANIPULATION COMMANDS

- a) **cat filename** : to view files. This will print out entire contents of file to terminal
eg: cat pokhara
- b) **touch filename** : to create an empty file with name given in current directory.
eg: touch infomax
- c) **nano filename**: to edit the contents of file. To save press CTRL + X. and enter.

GCC COMPILER

\$ gcc -o filename filename.c

\$./ filename

sudo apt-get update : to update ubuntu version

sudo apt-get install gcc : compiler for c and c++ Eg: \$ gcc -o add add.c \$./ add

OUTPUT

```
ankit@ankit-virtual-machine:~$ pwd
/home/ankit
ankit@ankit-virtual-machine:~$
```

```
ankit@ankit-virtual-machine:~$ ls
Desktop  Downloads  Pictures  snap      Videos
Documents Music      Public    Templates
ankit@ankit-virtual-machine:~$
```

```
ankit@ankit-virtual-machine:~$ mkdir ankit
ankit@ankit-virtual-machine:~$ ls
ankit  Documents  Music      Public  Templates
Desktop Downloads  Pictures  snap    Videos
ankit@ankit-virtual-machine:~$ cd ankit
ankit@ankit-virtual-machine:~/ankit$
```

```
ankit@ankit-virtual-machine:~$ ls
ankit Desktop Documents Downloads Music Pictures Public snap Templates Videos
ankit@ankit-virtual-machine:~$ rm -d ankit
ankit@ankit-virtual-machine:~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos
ankit@ankit-virtual-machine:~$
```

```
ankit@ankit-virtual-machine:~$ ls
ankit Desktop Downloads Pictures snap Videos
ankit.txt Documents Music Public Templates
ankit@ankit-virtual-machine:~$ cd ankit
ankit@ankit-virtual-machine:~/ankit$ touch ankit1.txt
ankit@ankit-virtual-machine:~/ankit$ nano ankit1.txt
ankit@ankit-virtual-machine:~/ankit$ cat ankit1.txt
Hello all! I am Ankit Pangen. This is lab 1 of Operating system.
ankit@ankit-virtual-machine:~/ankit$
```

```
ankit@ankit-virtual-machine:~/ankit$ touch add.c
ankit@ankit-virtual-machine:~/ankit$ nano add.c
ankit@ankit-virtual-machine:~/ankit$ cat add.c
#include <stdio.h>
int main()
{
int a=5,b=2;
printf("The sum of a+b is %d", a+b);
return 0;
}
```

```
ankit@ankit-virtual-machine:~/ankit$ sudo apt install gcc
```

```
ankit@ankit-virtual-machine:~/ankit$ gcc -o add add.c
ankit@ankit-virtual-machine:~/ankit$ ls
add  add.c  ankit1.txt
ankit@ankit-virtual-machine:~/ankit$ ./add
The sum of a+b is 7ankit@ankit-virtual-machine:~/ankit$
```

CONCLUSION

HENCE, WE STUDIED SOME BASIC LINUX COMMANDS AND IMPLEMENTED GCC COMPILER IN UBUNTU OPERATING SYSTEM IN LAB.